Supplementary materials:
Scenario audit of Cuba's 2021–2026 population decline

Yasset Pérez-Riverol

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Status. Model E is *retired*. It is documented here only for traceability: it was superseded by the age-standardised excess-mortality decomposition in the main paper, which estimates the same quantity from observed age denominators rather than a hand-weighted dimensionless index. Its figures are historical and are not current claims.

These materials accompany the main preprint. Models A–D are scenarios; Model D is an illustrative central path, not an identified estimator. Model E is provisional and is not preferred for public claims [Pérez-Riverol, 2026b,a].

1 Provenance of headline quantities

Table 1 classifies major inputs as observed (ONEI tables), derived (accounting identities), assumed (scenario priors), or expert-coded (HSDS features). Full machine-readable version: `data/provenance.yaml`.

Table 1: Provenance taxonomy (selected quantities).

Quantity	Class	Source	Status
ONEI B, D, 2022–25	observed	Anuario / Indicadores	canonical
Identity R	derived	`model_from_2021.py`	canonical
Scenario envelope 1.8–2.4 M	assumed	Models A–D priors	primary
Central scenario 8.58 M	assumed	central margins	scenario value
Destination floor 1.05 M	derived	`destinations_settled.csv`	ledger
Attributable mortality (2024–25 $\sim$50k)	derived	`attributable_mortality.py`	age-standardised
Crude 2019-CDR residual $\sim$77k	derived	`mortality_schedule_residual.py`	*superseded*
HSDS S_t	expert-coded	`hsds_features.csv`	Model E only
Model E 8.96 M	assumed	HSDS + migration bridge	provisional

2 Life expectancy series

Official life expectancy at birth peaked in 2011–13 (78.5 years) and fell to 77.7 in 2018–20 (Figure 1). An independent 2021 estimate places it near 71 years [Albizu-Campos Espiñeira, 2023, Brønnum-Hansen et al., 2023]. Life expectancy re-expresses the same mortality shock rather than providing independent corroboration of Model D; its magnitude is large even relative to COVID-era losses documented internationally [Aburto et al., 2022, Msemburi et al., 2023].

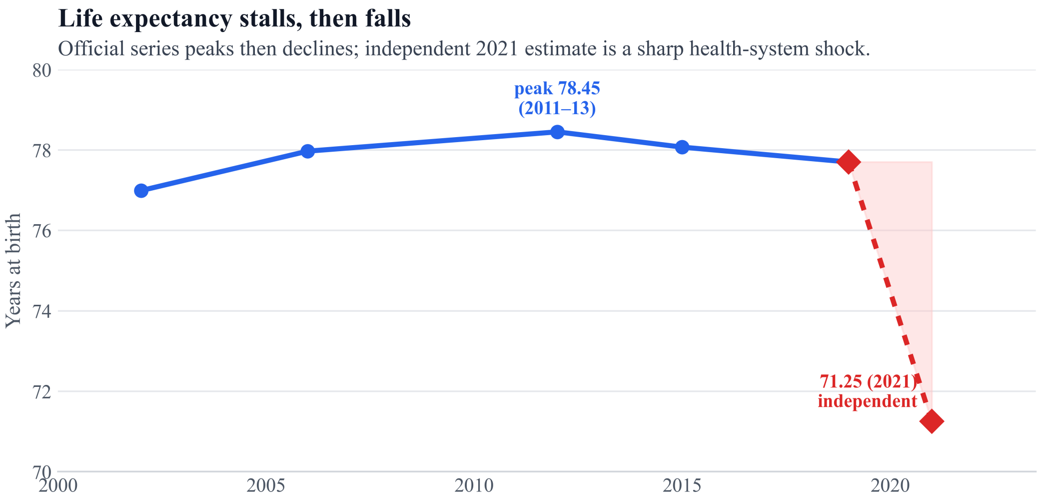

Figure 1: Life expectancy at birth: official triennial series and independent 2021 estimate.

3 Provincial intensity of exit

Subnational net outflow is national but uneven (Figure 2). ONEI *Indicadores 2025* reports Havana external net emigration near $-$**33 per 1,000** in 2025; western and central provinces empty faster than the east [ONEI, 2025]. If national migration accounting is biased, provincial *ranks* are more trustworthy than precise levels.

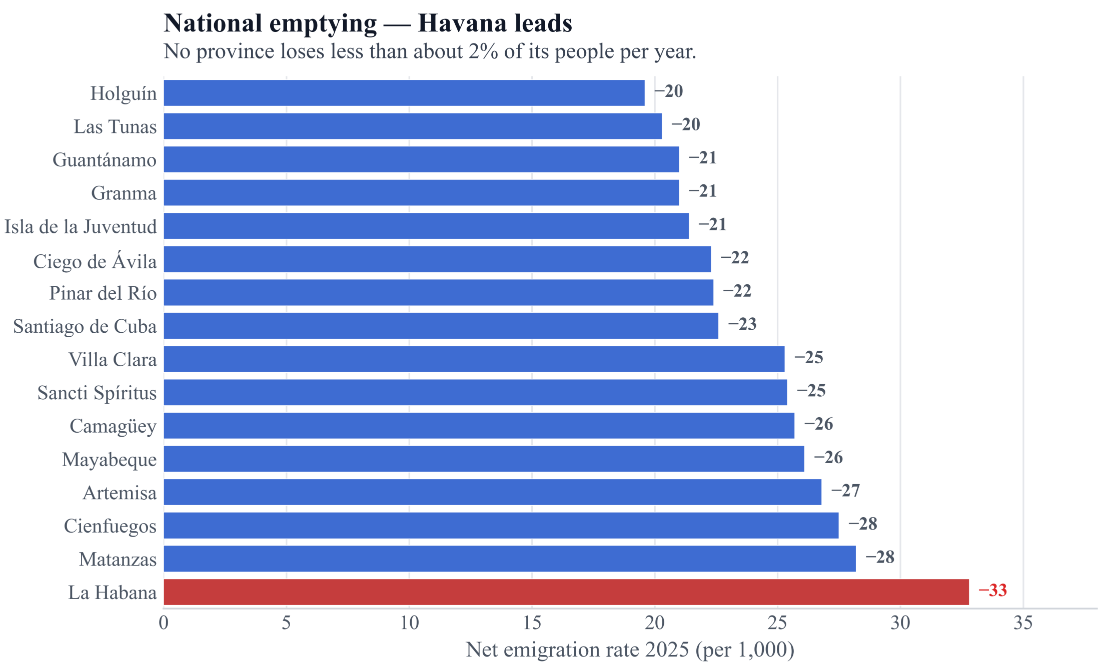

Figure 2: Net emigration rate by province, 2025 (ONEI-reported).

4 Model E: vital reconstruction (provisional)

Model E reconstructs deaths and births from a yearly health-system deterioration score S_t (ageing, health-system decadence, documented underreporting), then closes the identity with a Model D-like migration bridge. The spirit follows excess-mortality reconstruction under incomplete registration [Karlinsky and Kobak, 2021, Msemburi et al., 2023, Kishore et al., 2018], but uses a Cuban-specific score rather than a pandemic-only counterfactual.

Adversarial gate (why E is not promoted).

- Ageing enters both the expected-death schedule and the S_t bucket (double-count risk).
- "Ageing schedule" is currently a two-factor CDR proxy, not $\sum_a m_a P_a$.
- Model E's U_0 prior is much smaller than the central scenario's; the higher end-2025 stock ($\sim$8.96 M vs 8.58 M) is therefore partly a *different baseline prior*, not independent vital evidence.
- Expert-coded HSDS inputs (`hsds_features.csv`) use subjective feature weights; full sensitivity mapping is pending.
- Transfer from generic excess-mortality methods [Karlinsky and Kobak, 2021] to Cuban reporting institutions is not validated out of sample.
- P^* remains migration-identified under the reconstruction path.

Median end-2025 population about **8.96 million** (loss 2.13 million, 19.2% on the corrected base; migration share $\sim$88%). Figure 3 shows S_t and reconstructed versus registered vitals; Figure 4 places E beside A–D. Preferred public claims remain Model D until this gate is cleared [Pérez-Riverol, 2026b].

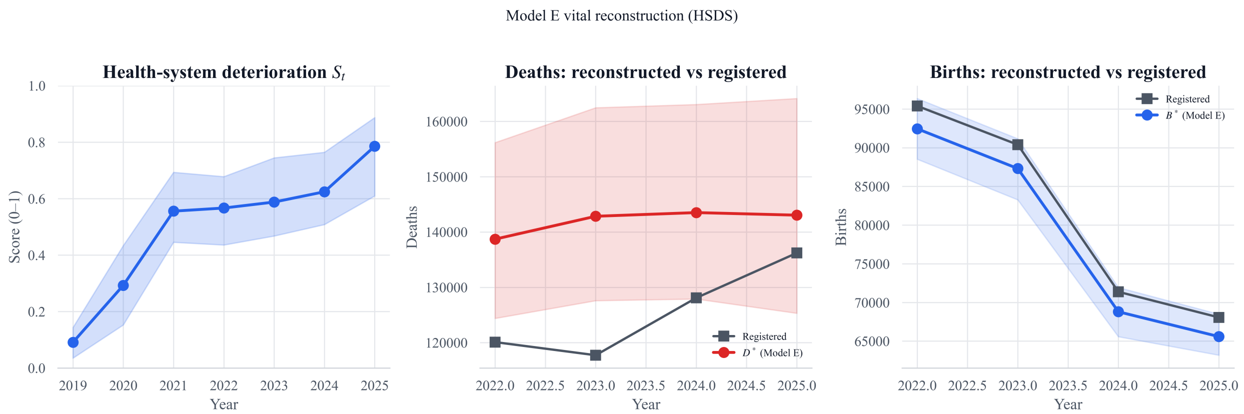

Figure 3: Health-system deterioration score S_t and Model E reconstructed deaths/births versus registered series (2022–2025).

Table 2: Models A–E medians (end-2025).

Model	Main assumption	Loss (M)	%	Pop. (M)
A · conservative	ONEI-anchored	1.80	16.4	9.16
B · crisis-adjusted	under-reg. + independent migration	2.19	20.4	8.57
C · upper stress	analog upper bound	2.41	22.6	8.26
Central (crisis-adjusted)	migration prior above ONEI + bounded D_{fac}	2.53	22.8	8.58
E · vital reconstr. (prov.)	HSDS $\rightarrow B^*, D^*$	2.13	19.2	8.96

5 Data pointers

Scripts and JSON artifacts for HSDS / Model E live under `demographics/scripts/hsds_*.py` and `demographics/artifacts/` in the companion repository [Pérez-Riverol, 2026b]. Adversarial notes: `artifacts/model_e_adversarial_notes.md`.

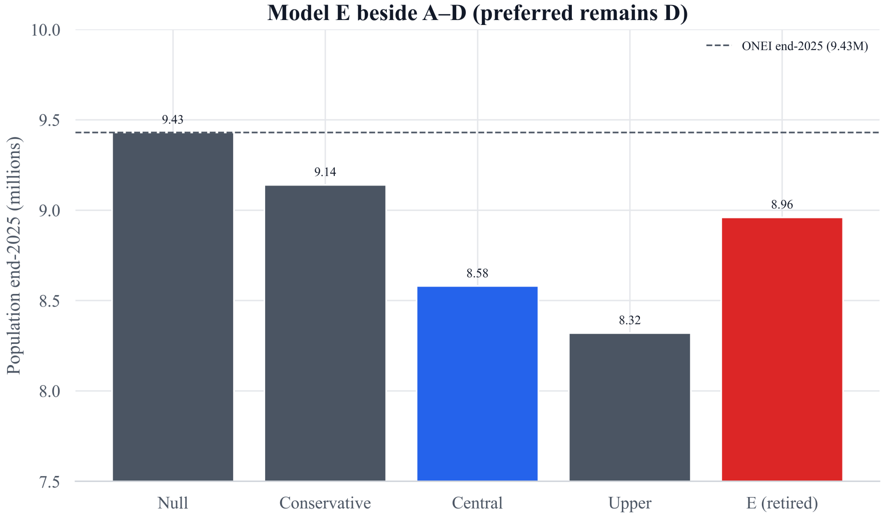

Figure 4: End-2025 population medians for Models A–E. Illustrative central scenario remains D (blue); E (red) is provisional.

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